

Assessment Task 2: Data exploration and preparation

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- Due Sunday by 23:59
- Points 100
- Submitting a file upload
- Available 27 Jul at 12:00 - 27 Sep at 23:59



Task details

This assessment will give you practical experience in data visualisation, exploration, and preparation (preprocessing and transformation) for data analytics. This assignment is individual work. Each of you will be working with an individual dataset that you can download from the link below.

Objectives:

- This assessment task addresses the following subject learning objectives (SLOs): 2 & 4
- This assessment task contributes to the development of the following Course Intended Learning Outcomes (CILOs): D.1

o

Scenario

Imagine a public health organization aiming to build an early detection health system to combat the growing incidence of diabetes across the population. To achieve this, researchers are leveraging nationwide health surveillance data collected through the Behavioral Risk Factor Surveillance System (BRFSS)—an annual survey tracking health behaviors, chronic conditions, and preventive care usage.

As you will see, this dataset is highly complicated and includes a lot of features that make this problem more challenging.

Your tasks include:

- o understanding the specifics of the dataset;
- o extracting information about each of the attributes, possible associations between them, and any other specifics of the dataset, that might be relevant to the business problems you developed in AT1.

The tasks in the assignment are specified below.

Datasets

For this dataset, you can access to the attributes headings from [here](https://studentutsedu-my.sharepoint.com/:f:/g/personal/maoying_qiao_uts_edu_au/lgBEJQbsZSPoRopj294H2J-IaUKsKXWtUmbfMluzDLkpuY?e=CQaGqB)  (https://studentutsedu-my.sharepoint.com/:f:/g/personal/maoying_qiao_uts_edu_au/lgBEJQbsZSPoRopj294H2J-IaUKsKXWtUmbfMluzDLkpuY?e=CQaGqB). Each student is assigned an individual table with the actual values of these attributes. You can download your dataset with names associated with your studentID using the link provided below.

Individual Student Datasets: [32130_AT2_individuals](https://studentutsedu-my.sharepoint.com/:f:/g/personal/maoying_qiao_uts_edu_au/lgBEJQbsZSPoRopj294H2J-IAR9-We3r6NofKz1MB_YO4VU?e=AhB4gf)  (https://studentutsedu-my.sharepoint.com/:f:/g/personal/maoying_qiao_uts_edu_au/lgBEJQbsZSPoRopj294H2J-IAR9-We3r6NofKz1MB_YO4VU?e=AhB4gf).

Tasks

1A. Initial data exploration

1. Identify the attribute type of each attribute in your dataset. If it's not clear, you may need to justify why you chose the type.
2. For each attribute, conduct below studies for each of them: Identify the values of the summarising properties for the attributes, including frequency, location and spread (e.g. value ranges of the attributes, frequency of values, distributions, medians, means, variances, percentiles, etc. - the statistics that have been covered in the lectures and materials given). Note that not all of these summary statistics will make sense for all the attribute types, so use your judgement! Where necessary, use proper visualisations for the corresponding statistics.
3. Using KNIME or Python, explore multiple attributes relationship of your dataset, and identify any outliers, clusters of similar instances, "interesting" attributes and specific values of those attributes. Note that you may need to 'temporarily' recode attributes to numeric or from numeric to nominal. The report should include the corresponding snapshots from the tools and an explanation of what has been identified there.

Present your findings in the assignment report.

1B. Data preprocessing

Perform each of the following data preparation tasks (each task applies to the original data) using your choice of tool:

1. Use the following binning techniques to smooth the values of the following attribute:
 - **MentHlth**
 - **PhysHlth**

For the attribute, you must apply:

- I. Equi-width binning
- II. Equi-depth binning

In the assignment report, for each of these techniques, you need to illustrate your steps. In your Excel workbook file place the results in separate columns in the corresponding spreadsheet. Use your judgement in choosing the appropriate number of bins - and justify this in the report.

2. Use the following techniques to normalise the following attribute:
 - **BMI**

For this attribute, you must apply:

- I. min-max normalization to transform the values onto the range [0.0-1.0].
- II. z-score normalization to transform the values.

The assignment report provides an explanation of each of the applied techniques. In your Excel workbook file place the results in separate columns in the corresponding spreadsheet.

3. Discretise the **age** attribute into the following categories:

- Young (ages 24 and younger)
- Early Adulthood (ages 25--34).
- early Middle Age (ages 35--44),
- late Middle Age (ages 45--64), and
- Late Adulthood (ages 65 and older).

Provide the frequency of each category in your dataset.

Your assignment report should provide an explanation of each of the applied techniques. In your Excel workbook file place the results in a separate column in the corresponding spreadsheet.

4. Binarise the **sex** variable [with values "0" or "1"].

Your assignment report should provide an explanation of the applied binarisation technique. In your Excel workbook file place the results in separate columns in the corresponding spreadsheet.

1C. Summary

At the end of the report include a summary section in which you summarise your findings. The summary **is not** a narrative of what you have done, but a condensed informative section of **what you have found** about the data that you should report to the Head of the Analytics Unit. The summary may include the most important findings (specific characteristics (or values) of some attributes, important information about the distributions, some clusters identified visually that you propose to examine, associations found that should be investigated more rigorously, etc.).

Deliverables

The deliverables are:

- **A report**, for which the structure should follow the tasks of the assignment, and
- **An Excel workbook** file with individual spreadsheets for each task (spreadsheets should be labelled according to the task names, for example, "1A"). Each of the results of parts (1) to (4) in task 1B should be presented in a separate sheet (and respectively table in the assignment report).

In the report, include a section (starting with a section title) for each of the tasks in the assignment.

Submission details

Length: The task requires the submission of a report (approx. 2500-3000 words or 18-25 pages using 11 or 12-point Times or Arial fonts). Please also upload the Excel spreadsheet containing your results. On average you will require

between 15 and 23 hours to complete this assignment.

Weight: 35%, marked out of 100

Criteria: see below Rubrics.

Submit:

- This assignment is individual work.
- Submit a report in Adobe PDF (preferable) or MS Word Doc with the filename (where xxxxxxxx is your student id):
 - report: "fda_a2_XXXXXXX.pdf" or "fda_a2_XXXXXXX.doc"
 - spreadsheet: "fda_a2_XXXXXXX.xls" or "fda_a2_XXXXXXX.xlsx"

Feedback: We plan to return marked assignments within 2 weeks of submission. Emails will be sent when marking is complete.

Academic Standards and Late Penalties: Please refer to [important guidance](https://canvas.uts.edu.au/courses/40826/pages/important-guidance) (<https://canvas.uts.edu.au/courses/40826/pages/important-guidance>).

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32130 Rubric: Assessment 2 (1)

Criteria	Ratings	Pts
Exploration of the dataset 1 focusing on individual attributes		

	30 Pts Full score [30] Exceeding the requirement in HD	27.75 Pts HD [25.5, 30) Correctly identifies all attribute types with well-considered justification when necessary. Correctly and imaginatively finding, explaining and justifying useful summary statistics for the attributes with appropriate and meaningful visualisations. A thorough exploration of the dataset identifying and characterising outliers and clusters of potentially attributes, with well-chosen visualisations (including clear descriptions and interpretation). Identification, description, and justification of interesting attributes.	24 Pts D [22.5,25.5) Correctly identifies all attribute types with well-considered justification when necessary. Correctly finding and explaining summary statistics for the attributes with appropriate visualisations. A good exploration of the dataset identifying and characterising outliers and clusters of potential attributes, with relevant visualisations (including clear descriptions). Identification and description of interesting attributes.	21 Pts C [19.5,22.5) Correctly identifies most attribute types with justification for all relevant attributes. Mostly correct finding and explaining of summary statistics for the attributes. Visualisations may be missing or inappropriate. Some identification and/or characterisation of outliers and clusters of potential attributes with visualisations (including some description but lacks indepth understanding). Some identification and/or description of interesting attributes.	17.25 Pts P [15,19.5) Correctly identifies most attribute types, some justifications may be missing. Mostly correct summary statistics for the attributes but not well explained. Limited use of appropriate visualisations. A bit of exploration, but not really coming to terms with outliers and clusters of attributes and minimal or no relevant visualisations. Poor identification and/or description of interesting attributes.	7.49 Pts F [0,15) Incorrect or limited identification of attribute types with no relevant justifications. Incorrect or limited summary statistics for the attributes lacking explanation. Poor or no use of appropriate visualisations. Limited or no exploration of the dataset with no consideration of outliers and clusters of attributes. Limited or no relevant visualisations. Incorrect or limited identification and/or characterisation of interesting attributes.	30 pts

Exploration of the dataset 2 focusing on multiple (≥ 2) attributes and overall exploration	30 Pts Full score [30] Exceeding the requirement in HD	27.75 Pts HD [25.5, 30] A thorough exploration of the dataset identifying and characterising outliers and clusters of data points, and potentially attributes, with well-chosen visualisations (including clear descriptions and interpretation). Identification, description, and justification of interesting attribute relationships.	24 Pts D [22.5,25.5) A good exploration of the dataset identifying and characterising outliers and clusters of data points, with relevant visualisations (including clear descriptions). Identification and description of interesting attribute relationships.	21 Pts C [19.5,22.5) Some exploration of the dataset, including multiple attributes but missing key information. Some identification and/or characterisation of outliers and clusters with visualisations (including some description but lacks indepth understanding). Some identification and/or description of interesting attribute relationships.	17.25 Pts P [15,19.5) A bit of exploration, but not really coming to terms with multiple attributes taken together and minimal or no relevant visualisations. Poor identification and/or characterisation of outliers and clusters.	7.49 Pts F [0,15) Limited or no exploration of the dataset with no consideration of multiple attributes. Incorrect or limited identification and/or characterisation of outliers and clusters. Limited or no relevant visualisations.	30 pts

Preprocessing of the dataset	20 Pts Full Score [20] Exceeding the requirement in HD	18.5 Pts HD [17, 20] Correct, well explained and justified preprocessing carried out using all relevant techniques.	16 Pts D [15,17] Correct and justified preprocessing carried out using all relevant techniques.	14 Pts C [13, 15] Mostly correct preprocessing, but not well explained or justified only using some relevant techniques.	11.5 Pts P [10, 13] Mostly correct preprocessing, but no explanation or justification using some relevant techniques.	5 Pts F [0,9.99] Poor or no preprocessing, with no explanation or justification or use of relevant techniques	20 pts
Summary of findings	20 Pts Full Score [20] Exceeding the requirement in HD	18.5 Pts HD [17, 20] Insightful concise summary of results and meaning written with careful thought towards the audience. Language use is error free and the report extremely well presented and organised.	16 Pts D [15,17] Clear summary written with reference to results and/or meaning. Some thought towards the audience. Language use is mostly error free and the report well presented and organised.	14 Pts C [13, 15] Reasonable summary but with some reference to results and/or meaning. Not much thought about the audience. Language use is mostly acceptable with some areas of improvement needed regarding presentation or organisation.	11.5 Pts P [10, 13] Summary mostly just says what exploration was done, not the results or meaning. Limited thought of the audience. Language use may be vague with many aspects needing improvement.	5 Pts F [0,9.99] Poor or unclear summary lacking reference to results or meaning. Poor or no thought of the audience. Language use is poor and hard to understand.	20 pts
Total points: 100							